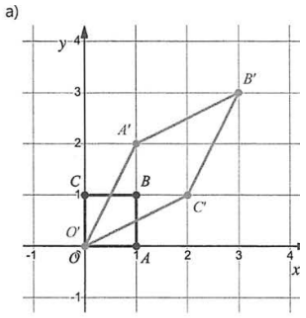
# 2D Matrices as Linear Transformations

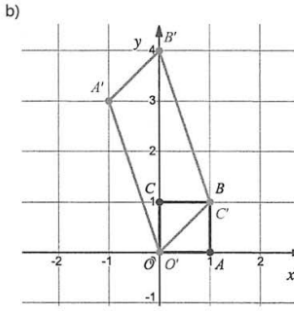
Questions 1–22

**Question 1.** The diagrams below show the unit square, $OABC$, and its image, $O'A'B'C'$, after a transformation. In each case, write down the matrix that represents this transformation:

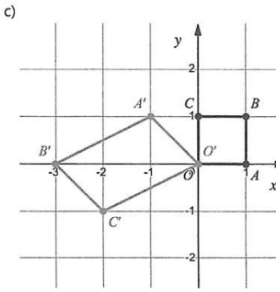
**a)**




**b)**

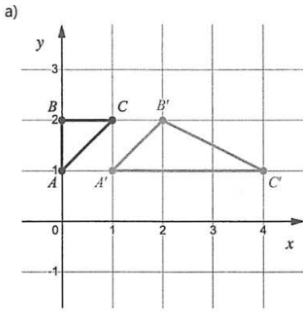



**c)**

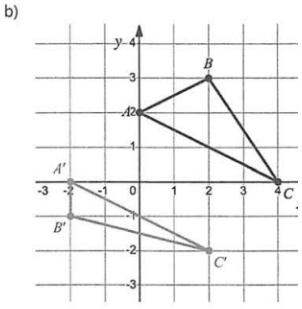



**Question 2.** The diagrams below show the triangle $ABC$ and its image, $A'B'C'$, after a transformation. In each case, write down the matrix that represents this transformation:

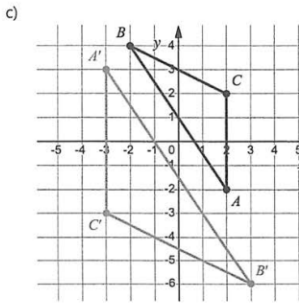
**a)**




**b)**




**c)**



**Question 3.** Find the image of the coordinates $(2, 3)$ after the following linear transformations:

a) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

b) $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$

c) $\begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$

d) $\begin{pmatrix} 1 & 0 \\ 5 & 1 \end{pmatrix}$

e) $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$

f) $\begin{pmatrix} -1 & 2 \\ 3 & -4 \end{pmatrix}$

**Question 4.** A shape is transformed by the following matrices. Describe what happens in each case:

a) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

b) $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

**Question 5.** Find the image of the coordinates $A(-3,-5)$ and $B(2,-1)$ after the following linear transformations:

a) $\begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$

b) $\begin{pmatrix} -3 & 1 \\ -4 & 2 \end{pmatrix}$

c) $\begin{pmatrix} 1.2 & 3.6 \\ -2.8 & -1.9 \end{pmatrix}$

d) $\begin{pmatrix} 0 & 0 \\ 0 & 7.7 \end{pmatrix}$

**Question 6.** The matrix $\mathbf{A} = \begin{pmatrix} 5 & -3 \\ 4 & -8 \end{pmatrix}$ represents the transformation $T$ which transforms the points $P(1,2)$ and $Q(-3,4)$ to the points $P'$ and $Q'$.

a) Find the distance $PQ$.

b) Find the distance $P'Q'$.

**Question 7.** Find the $2 \times 2$ matrix that maps $(4, -1)$ and $(-2, -3)$ onto $(7, 1)$ and $(-7, -11)$ respectively.

**Question 8.** Find the $2 \times 2$ matrix that maps $(5, 6)$ and $(2, -1)$ onto $(-37, 50)$ and $(-8, 3)$ respectively.

**Question 9.** Find the $2 \times 2$ matrix that swaps the coordinates $(1, 2)$ and $(1, 1)$.

**Question 10.** Given the matrix $\begin{pmatrix} 6 & 8 \\ -2 & 9 \end{pmatrix}$ maps $(a, b)$ onto $(46, 78)$, find the values of $a$ and $b$.

**Question 11.** Given the matrix $\begin{pmatrix} 4 & 10 \\ 3 & 2 \end{pmatrix}$ maps $(a, b)$ and $(c, d)$ onto $(8, -5)$ and $(76, 13)$ respectively, find the values of $a$, $b$, $c$ and $d$.

**Question 12.** Given the matrix $\begin{pmatrix} 3 & k \\ -4 & 2k \end{pmatrix}$ maps $(a, 16)$ onto $(-23, -236)$, find the values of $k$ and $a$.

**Question 13.** Given the matrix $\begin{pmatrix} 4k & -5k \\ 8k & 3k \end{pmatrix}$ maps $(k, 8)$ onto $(-52k, 0)$, where $k$ is nonzero, find the value of $k$.

**Question 14.** Given the matrix $\begin{pmatrix} k+3 & k-5 \\ k-7 & k+1 \end{pmatrix}$ maps $(3k, k-6)$ onto $(44k+8, 17k+5)$, find the value of $k$.

**Question 15.** Given the matrix $\begin{pmatrix} 3 & 2 \\ -1 & 1 \end{pmatrix}$ maps $P(1, k)$ and $Q(k, 1)$ to the points $P'$ and $Q'$, where the distance between $P'$ and $Q'$ is $2\sqrt{5}$ units, find the possible values of $k$.

**Question 16.** Given the matrix $\begin{pmatrix} 8 & -6 \\ 4 & -3 \end{pmatrix}$ maps any point onto the line $l$, find the equation of the line $l$.

**Question 17.** Given the matrix $\begin{pmatrix} 1 & -3 \\ 2 & 1 \end{pmatrix}$ maps the line $y = x$ onto the line with equation $y = ax$, find the value of $a$.

**Question 18.** Given the matrix $\begin{pmatrix} -2 & -5 \\ 8 & -3 \end{pmatrix}$ maps the line $y = x - 2$ onto the line with equation $y = ax + b$, find the values of $a$ and $b$.

**Question 19.** Given the matrix $\mathbf{A} = \begin{pmatrix} a & 3 \\ 2 & b \end{pmatrix}$ maps the line $y = 2x + 1$ onto the line with equation $y = 3x + 2$, find the values of $a$ and $b$.

**Question 20.** Given the matrix $\mathbf{A} = \begin{pmatrix} 0 & -3 \\ 3 & 0 \end{pmatrix}$ maps the circle $C$ with equation $(x - 4)^2 + y^2 = 8$ onto the circle $C'$, find the equation of $C'$.

**Question 21.** Given the matrix $\begin{pmatrix} -2 & 0 \\ 0 & -3 \end{pmatrix}$ maps the line $y = x^2 + 2x$ onto the line with equation $y = ax^2 + bx + c$, find the values of $a$, $b$ and $c$.

**Question 22.** Given the matrix $\begin{pmatrix} -2 & -2 \\ 6 & -6 \end{pmatrix}$ maps the curve $y = \frac{1}{x}$ onto the curve with equation $ax^2 - by^2 = 1$, find the values of $a$ and $b$.